Teaching Dossier

Alain Maubert Crotte

Portfolio: alainmaubert.github.io/portfolio/

Teaching Evidence: alainmaubert.github.io/portfolio/teaching/

1. Statement of Teaching Philosophy

My teaching philosophy has been shaped by an academic and professional journey spanning Mexico, China, and Canada. At the University of Regina, my teaching has unfolded across three distinct faculties: the Faculty of Science, the Faculty of Media, Art, and Performance, and the Faculty of Arts.

I believe that technology's deepest value emerges when it serves as a medium for creativity, culture, and inclusion. Teaching is not simply about software proficiency; it is about empowering diverse students to become innovative creators and socially responsible technologists.

- **Interdisciplinary Collaboration:** I bridge disciplinary languages. When teaching with specialized hardware, such as VR headsets or Arduino kits, a student occasionally arrives without a required component. I maintain momentum by fostering peer support, encouraging classmates to share missing parts, and forming balanced teams where each student has a specific role (operating equipment, developing assets, documenting processes). This ensures learning continues uninterrupted.
- **Inclusion and Social Engagement:** My doctoral research on left-handed inclusion in computer games informs my pedagogy. I encourage students to critically examine who technologies serve, how interfaces include or exclude, and how creative technologies can promote social justice.
- **Process over Polish:** In production courses, this means short demonstrations followed by studio time, frequent small experiments rather than a single high-stakes assignment, and documentation of iterations including the ones that failed.

Read the full Teaching Philosophy (PDF)

2. Inventory of Teaching Responsibilities

I have maintained a continuous teaching record at the University of Regina since 2016, delivering courses in person and synchronously online via Zoom.

Sessional Lecturer

- **MAP 102: Exploring Cultural Regina** (Spring/Summer 2026). Experiential learning course involving site visits to cultural institutions across Regina.
- **CTCH / DES 312: Introduction to Computer Game and VR Design** (Winter 2024 to Winter 2026, five consecutive terms). Course development, lectures, and studio time covering the full production pipeline from concept to implementation using Unity.
- **CS 100: Introduction to Computers** (Fall 2023). Foundational concepts including computer organization, data representation, algorithms, and productivity applications. Covers the societal impact of computing, including privacy and security.
- **CS 408: Animation Software Design** (Fall 2022). Delivered fully online. Design and implementation of software for creating animations, covering motion control, keyframe-based animation, kinematics, and physically based animation.
- **CS 476: Software Development Project** (Winter 2022, Spring/Summer 2022). Project-based software architecture. Students work through the software development life cycle (SDLC), using UML for specification and design, and development frameworks for implementation.
- **CS 215: Web and Database Programming** (Fall 2016, Winter 2017, Spring/Summer 2016). Design and implementation of interactive database-driven web applications, focusing on client-server architectures, interface design, event-driven programming, and data modeling.

Lab Instructor

- **CTCH 223 / CS 207: Building Interactive Gadgets** (Winter 2025, Fall 2025, Fall 2026). Introduction to building and controlling interactive devices. Covers physical computing, Arduino hardware, sensors, and actuators.
- **CS 210: Data Structures and Abstractions** (Fall 2021, Winter 2022). Lab instruction covering C++ programming concepts: linked lists, operator overloading, recursion, stacks, queues, hash tables, and binary search trees.

3. Evidence of Teaching Effectiveness

My teaching effectiveness is evidenced by consistent reappointment across multiple faculties, positive student evaluations, and positive student outcomes.

- **Student Evaluations:** In **CTCH / DES 312**, student evaluations averaged **4.3 out of 5.0**. Students consistently highlighted my approachability, respect for students, and ability to make complex topics accessible. In upper-level project courses like **CS 476**, evaluations averaged **3.6 out of 4.0** for preparation and communication.
- **Selected Student Feedback:**
  - *"Friendly, approachable, respectful, and capable instructor. Course is enjoyable and useful. Thank you!"* (CTCH / DES 312)
  - *"The instructor shows live examples of various code as well as present fun activities which are helpful in learning the coding concepts in a very easy manner."* (CS 215)
- **Reflective Practice and Continuous Improvement:** Early in my teaching career (CS 100, CS 215), feedback indicated that the dense volume of material sometimes felt challenging to absorb within the class period. I have since heavily revised my approach, shifting from dense lectures toward live coding demonstrations and interactive problem-solving. When students in CTCH / DES 312 struggled with the final project, I provided supplementary Unity documentation and extended deadlines when necessary to ensure comprehension over strict pacing.

View the full summary of teaching evaluations (PDF)

Student Outcomes: Examples of student work, including game design projects, VR prototypes, and physical computing labs, can be viewed on my online portfolio:

- Student Game Design Projects
- Immersive VR Interaction
- Physical and Embedded Interaction

4. Professional Development

I am committed to reflective, evidence-based teaching practices.

- **Graduate Student Teaching Development Program (URGSTDP):** Completed the University of Regina certificate program. Key training modules included:
  - *Anti-Oppressive Approaches*
  - *Indigenizing Curriculum*
  - *Identifying Learning Outcomes, Assessment, and Evaluation*
  - *Evidence-based Teaching and Learning Strategies*
  - *Untraditional Learning Methods / Student Centered Learning*
- **Technical and Creative Practice:** During my time as Head of Mixed Reality at the ReACT Lab, I developed research in VR, AR, and 3D modeling, including the design and development of the VisionAR: Impairment Simulator and the VR Regalia: Museum Interaction Prototype. I bring this applied industry and research experience into the classroom to inform my teaching across Creative Technologies (CTCH) and the Faculty of Media, Art, and Performance (MAP).

5. Scholarship of Teaching and Learning

My approach to teaching goes beyond the classroom to investigate how students learn effectively and how technology can enhance that process.

- **Research-Informed Pedagogy:** My doctoral research involves conducting controlled usability studies with 100 participants using validated instruments (such as PENS and Fitts's Law tests). I apply this rigorous, evidence-based approach to understanding how students interact with educational software and hardware in my classrooms.
- **Innovative Curriculum Design:** Adapting **MAP 102: Exploring Cultural Regina** was an exercise in testing experiential learning models. By moving the classroom into the city and partnering with cultural institutions, I evaluated how community-based learning impacts student engagement and critical thinking.
- **Iterative Assessment Design:** In CTCH / DES 312, I continuously experiment with assessment structures (such as shifting from high-stakes exams to iterative project rubrics) to better measure practical skill acquisition and creative problem-solving.
- **Accessible and Immersive Tool Design:** My research extends beyond the classroom into the design of interactive systems. I developed VisionAR: Impairment Simulator, a web-based accessibility tool for desktop and mobile AR that simulates visual conditions like macular degeneration and tunnel vision. I also designed the VR Regalia: Museum Interaction Prototype for Meta Quest 3, exploring spatial interaction and haptic feedback for cultural heritage display. These projects serve as case studies in my courses for designing inclusive, user-centered experiences.

Interactive Research Artifact: To demonstrate how I apply usability testing to my own design, you can play the game I built from scratch in Unity as the research instrument for my doctoral studies:

- Play MegaByte Meltdown (WebGL build)

6. Educational Leadership and Curriculum Development

I actively contribute to curriculum design and student success beyond my immediate classroom duties.

- **Course Design:** I adapted **MAP 102: Exploring Cultural Regina** to strengthen its experiential learning components. This involved coordinating with external cultural partners and refining the curriculum to focus on critical engagement with the city's arts organizations.
- **Curriculum Development:** I contributed directly to the structure, assignments, evaluation criteria, and rubrics for **CTCH / DES 312**.
- **Student Mentorship:** I support student learning beyond the classroom by encouraging participation in industry events like the Global Game Jam (featured in Global News) and Skills Canada. I also provide guidance on portfolio development and hardware troubleshooting in CTCH 223 to ensure students with varying technical backgrounds can succeed.